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From Noisy Labels to Class-Aware Models: An Integrated Strategy for Maximizing Neutral-Class Recall Without Sacrificing Overall Accuracy

Foundational Data Cleaning with Confident Learning

Confident Learning (CL) emerges as a critical, algorithmic-first approach to addressing the challenge of improving neutral-class recall without sacrificing overall accuracy. Its core function is not to directly boost minority-class performance but to enhance the underlying quality of the training data by systematically identifying and correcting label errors [3]. In datasets where the neutral class is underrepresented and potentially ambiguous, mislabeled instances are particularly detrimental. A sample from the positive class incorrectly labeled as neutral dilutes the signal for the neutral class, making it harder for the model to learn its true characteristics and thereby lowering its recall. CL directly confronts this issue by operating on the principle that a correctly specified model will almost never be simultaneously correct and highly confident in its prediction for a mislabeled example [4,6,9]. By flagging such discrepancies, CL provides a powerful tool for cleaning the dataset before the model is even trained, which serves as a foundational step for any subsequent advanced strategy. The process involves generating predictions and then searching for instances where the model exhibits high confidence in a predicted class that differs from the provided ground-truth label; these flagged instances are considered strong candidates for being mislabeled [2].

The practical application of Confident Learning has demonstrated tangible benefits across various domains. For instance, in medical image segmentation, a study assessed the PBNR method's ability to improve accuracy by removing high-noise label instances, showcasing the efficacy of targeting noisy labels for correction [2]. The framework extends beyond simple removal; it can generate sample-specific weights, allowing the model to pay less attention to potentially erroneous examples during training rather than discarding them entirely [1]. This nuanced approach helps preserve valuable information while mitigating the negative impact of incorrect labels. The importance of this technique is underscored by the prevalence of noisy labels in real-world datasets, which can

severely degrade learning performance if left unaddressed [116275]. Research indicates that models trained on data with noisy labels perform poorly when tested on clean data, highlighting the necessity of a robust noise-handling mechanism during the training phase [470471]. By integrating CL as a pre-processing step, one ensures that the model learns from a higher-quality signal, which is a prerequisite for achieving balanced performance across all classes, including the challenging neutral class. The joint estimation of label errors and model uncertainty is a key strength of CL, making it a principled method for tackling imperfectly annotated data [3].

From an implementation perspective, applying Confident Learning involves several steps. First, an initial model is trained on the full dataset. Then, the algorithm analyzes the model's confidence scores (typically the maximum probability assigned to any single class for a given input) against the ground-truth labels. Instances that meet the criteria for likely label errors are identified. A crucial aspect of this process is the need for human-in-the-loop validation. While CL is a powerful heuristic, it is not infallible and may flag legitimate but difficult-to-classify examples as errors. Therefore, having an expert review the flagged samples is essential before any corrective action is taken [201]. This verification step prevents the accidental deletion of rare but valid instances from the minority class, which could further harm recall. After validation, the options include deleting the sample, correcting its label, or down-weighting its influence in the loss function. Once the dataset is cleaned, the entire modeling process is repeated from scratch. This iterative refinement loop can continue until no more significant label errors are detected or until performance plateaus. The computational overhead of this process is generally considered acceptable, especially when compared to the gains in model robustness and accuracy [6]. By starting with a cleaner dataset, the base model is more likely to achieve a better balance between precision and recall, setting a stronger foundation upon which more sophisticated techniques like soft labeling or ordinal regression can be applied to specifically target the neutral class.

Nuanced Learning through Soft Labeling

Soft labeling represents a paradigm shift from the rigid, deterministic nature of traditional "hard" labels, offering a more nuanced approach to handling ambiguity in classification tasks. Instead of assigning a single class to each data point, soft labeling assigns a probability distribution across all possible classes, reflecting the degree of certainty or the consensus among multiple sources [19 251]. This method is particularly well-suited for subjective tasks like sentiment analysis, where the boundaries between

classes, especially for the "neutral" class, can be blurry. Human annotators often disagree on the classification of borderline cases, and soft labeling provides a formal way to capture this disagreement within the training data itself [19]. By training a model on these probabilistic targets, rather than forcing it to learn from a single, potentially arbitrary choice, the model is exposed to a richer and more realistic representation of the data's complexity. This can lead to improved robustness and better generalization, as the model learns to handle uncertainty rather than overfitting to potentially noisy hard labels.

The evidence supporting the efficacy of soft labeling is compelling. One study focused on stance detection, a highly subjective task, created soft labels by converting discrete votes from multiple annotators into continuous values using a softmax function [19]. The results showed that a classifier trained on this multi-perspective approach with soft labels outperformed a baseline model trained on majority-vote hard labels, achieving higher F1-scores [19]. This improvement underscores the value of incorporating diverse perspectives during training. The theoretical underpinning for this approach lies in the use of a corresponding "soft loss" function, typically categorical cross-entropy, which measures the divergence between the model's output distribution and the soft label distribution [19] [106]. This allows the model to be penalized not just for predicting the wrong class, but for assigning the wrong *degree* of probability to each class, encouraging a more calibrated and accurate set of predictions. Furthermore, soft labels can be generated from various sources, including crowdsourced annotations, where the frequency of votes for each class forms the basis of the distribution [252], or even from the outputs of Large Language Models, although this requires careful consideration of the LLM's calibration to avoid propagating biases [477].

However, there are important considerations and potential trade-offs associated with soft labeling. One noted consequence is that models trained on soft labels may exhibit lower confidence scores on average [19]. This is not necessarily a drawback but rather a reflection of the inherent uncertainty captured in the training data. It signifies that the model has learned to recognize instances that are genuinely ambiguous and avoids making overconfident, and potentially incorrect, classifications. This aligns with the broader goal of creating reliable classifiers whose confidence scores faithfully reflect evidential strength [220]. Another challenge is the generation of high-quality soft labels. When relying on multiple human annotators, inter-rater agreement can be low, requiring a sufficient number of annotators to create a stable distribution [19]. When using automated methods, such as LLMs, the quality of the generated probabilities becomes paramount. Uncalibrated LLM distributions have been shown to propagate bias into the training data, undermining the potential benefits of the approach [477]. Therefore, the successful implementation of soft labeling hinges on two factors: first, the existence of a

source of uncertainty to represent (e.g., human disagreement or intrinsic data ambiguity), and second, the availability of a reliable method for generating those probability distributions. If implemented correctly, soft labeling offers a direct and elegant solution to the problem of improving neutral-class recall by teaching the model what "in-between" looks like, rather than forcing every ambiguous case into a binary choice.

Structural Optimization via Ordinal Regression

Ordinal regression presents a fundamentally different and potentially powerful approach to the classification problem by leveraging the inherent order of the classes. Unlike standard classification, which treats categories like "positive," "neutral," and "negative" as nominal and independent, ordinal regression recognizes that these classes exist on a meaningful continuum [405][406]. This structural assumption allows the model to make more informed predictions by constraining its decision boundaries. For instance, the model learns that predicting "positive" for an instance that is actually "neutral" is a smaller error than predicting "strongly positive," because the former is adjacent on the ordinal scale while the latter is not. This implicit knowledge about class relationships can lead to more robust models, especially in domains like sentiment analysis where opinions naturally exist on a spectrum [56][333]. The goal is to model the cumulative probability of an instance belonging to a certain class or any class below it in the hierarchy [40].

The methodology for implementing ordinal regression is flexible. One common approach is to frame the K-class ordinal problem as a series of K-1 binary classification tasks, where each task distinguishes between "class i or lower" versus "class i+1 or higher" [52]. This decomposition simplifies the problem and allows for the use of standard binary classifiers within an ordinal framework. Alternatively, specialized models can be developed that explicitly account for the ordinal nature of the data, often by incorporating a monotonicity constraint that ensures the probability of belonging to a higher class decreases as the input features move in a direction that would suggest a lower class [448]. This approach has found success in various applications, including risk assessment, where outcomes are often measured on an ordinal scale, and emotion recognition, where emotional valence and arousal can be treated as ordered categories [76][138][271]. Studies have shown that ordinal models can achieve greater accuracy and more accurately recover observed category proportions compared to their non-ordinal counterparts, demonstrating the value of incorporating domain knowledge about class structure [37].

Despite its theoretical advantages, the effectiveness of ordinal regression is conditional and not guaranteed to outperform standard classification in all scenarios. A critical finding from one study revealed that an ordinal regression variant actually underperformed a standard classification model across all evaluated metrics, despite utilizing richer ranking information [45]. This highlights a crucial point: the assumption of a meaningful ordinal relationship must be validated for the specific dataset and task at hand. If the class ordering is weak, arbitrary, or not supported by the data patterns, imposing this structure can introduce bias and hinder performance. The choice between different modeling frameworks, such as different types of ordinal regression models, also involves trade-offs, for example, between the efficiency of some models and the interpretability of others [44]. To apply this method effectively, one must first statistically validate the assumption of an ordinal relationship, perhaps using measures like Cohen's Kappa statistic [325]. If the assumption holds, benchmarking an ordinal model against a state-of-the-art classifier (like a tuned XGBoost or Transformer) is essential. Evaluation should focus on metrics appropriate for ordinal data, such as standard accuracy, Mean Absolute Error (MAE) which measures the average distance between predicted and true ranks, and "Off-by-1" accuracy, which might be more forgiving in this context [43] [326]. Only through rigorous testing can one determine if the added structural complexity of ordinal regression translates into the desired improvements in both overall accuracy and neutral-class recall.

Scalable Annotation Enhancement with Large Language Models

The application of Large Language Models (LLMs) for re-labeling and data enhancement offers a pathway to scalable and cost-effective solutions, but it comes with significant caveats that require careful management. LLMs possess deep contextual understanding, enabling them to perform complex annotation tasks that were previously labor-intensive [53]. They can be employed to pre-annotate large datasets, augment existing data to address class imbalance, or even generate new, synthetic training examples [496]. One notable application involves a semi-automated pipeline that reduces manual labeling effort to as little as 10% by using high-confidence LLM-based pre-annotations, which are then verified [66] [151]. This scalability is a major advantage, as it allows for the creation of much larger and more diverse training sets. Furthermore, empirical studies have shown that in some contexts, LLM-generated labels can surpass the quality of original human annotations, particularly for high-confidence predictions when measured against expert

re-labeling [147]. This suggests that LLMs can act as powerful "weak labelers," providing a rich source of training signals that should be treated as constraints rather than absolute ground truth [4].

However, the deployment of LLMs for re-labeling is fraught with challenges related to reliability, consistency, and bias. A primary concern is that the probability distributions output by LLMs are often uncalibrated, meaning they do not perfectly reflect the model's actual accuracy. Using these uncalibrated distributions as soft labels can inadvertently propagate and amplify existing biases into the training data, leading to a degradation in model fairness and performance [477]. Model performance is also highly sensitive to prompt design, where subtle changes in phrasing can lead to significantly different outputs [113]. Moreover, LLMs are known to exhibit behaviors like sycophancy, where they attempt to please the user and provide misleading answers, and meta-cognitive failures, where they form deceptive internal representations [71] [191]. These issues mean that a naive approach of simply replacing human labels with LLM labels is risky and likely to fail. Rigorous evaluation and a structured verification framework are therefore non-negotiable prerequisites for any LLM-driven re-labeling initiative.

Given these complexities, a more prudent and effective strategy involves using LLMs in a targeted and controlled manner. Rather than attempting to fix every label in the original dataset, which is a high-risk endeavor, LLMs are best utilized for specific, lower-risk tasks. One of the most promising applications is data augmentation. By generating synthetic text that falls clearly into the positive or negative sentiment categories, LLMs can be used to balance the dataset and provide more robust examples for the model to learn from, without needing to precisely re-label ambiguous neutral instances [446]. Another safe application is high-confidence pre-annotation, where the LLM generates labels for the entire dataset, and a subsequent filtering step retains only those predictions that meet a high-confidence threshold [401][402]. This filtered subset can then be reviewed by humans, drastically reducing the manual effort required. A hybrid human-LLM collaborative approach, where the LLM generates labels and explanations that are then assessed by a verifier, has also been proposed as a viable path forward [201]. Ultimately, the integration of LLMs into the labeling process should be viewed as a tool to assist and scale human expertise, not replace it. Any resulting model must undergo rigorous evaluation using the dual-metric objective of maintaining >96% overall accuracy while maximizing neutral-class recall to ensure that the benefits of automation do not come at the cost of performance on the target class.

Comparative Analysis and Integrated Workflow

The four advanced strategies—Confident Learning, Soft Labeling, Ordinal Regression, and LLM-based Re-labeling—are not mutually exclusive but rather complementary techniques that can be integrated into a systematic, multi-stage workflow to address the dual objectives of improving neutral-class recall and maintaining high overall accuracy. Each method operates at a different level of abstraction, from foundational data cleaning to structural optimization and scalable enhancement. A comparative analysis reveals their distinct roles and relative strengths, guiding the formulation of an optimal integrated approach.

Strategy	Primary Mechanism	Key Benefit for Neutral Class	Risk/Challenge	Best Applied At
Confident Learning	Identifies and removes/corrects label errors in the training set.	Improves recall by ensuring the model learns from accurate examples, preventing confusion caused by mislabeled neutral instances.	Potential for false positives; may remove rare but valid minority-class examples if not carefully managed.	**Pre-processing:** As a foundational first step to clean the dataset before any model training begins.
Soft Labeling	Replaces hard labels with probability distributions, capturing ambiguity and human disagreement.	Directly teaches the model the semantics of "in-betweenness," making it more adept at recognizing borderline neutral cases.	Can lead to lower model confidence scores; requires a reliable method for generating high-quality probability distributions.	**Training:** During model training, using a soft-loss function to leverage nuanced label information.
Ordinal Regression	Models the inherent order of classes (e.g., Negative < Neutral < Positive).	Improves generalization by constraining predictions to be structurally plausible, reinforcing the intermediate nature of the neutral class.	Performance is conditional on the validity of the ordinal assumption; may underperform standard classification if the assumption is invalid.	**Model Selection:** As an alternative modeling paradigm if a clear ordinal relationship between classes can be established.
LLM Re-labeling	Uses LLMs for scalable pre-annotation, data augmentation, or synthetic data generation.	Augments the dataset with more examples, helping to balance class distributions and reinforce class boundaries.	High risk of propagating LLM biases and errors; requires careful prompting, calibration, and human verification.	**Augmentation:** Primarily for data augmentation or high-confidence pre-annotation, not wholesale label replacement.

An integrated workflow that leverages the strengths of each stage logically progresses from data purification to nuanced learning and final enhancement. The recommended sequence begins with **Phase 1: Data Foundation & Cleaning using Confident Learning**. Applying CL first purifies the dataset by removing the most obvious sources of noise, such as grossly mislabeled examples. This step is critical because all subsequent strategies will be working with a higher-quality signal, which is fundamental to achieving balanced performance. After cleaning the data, with human oversight to prevent the loss of valid minority-class instances, the process moves to **Phase 2: Nuanced Learning with Soft Labeling**. With a cleaner dataset, soft labels can be generated from remaining

human annotator disagreements or, cautiously, from a well-calibrated LLM. Training the model with these soft labels provides it with a direct lesson in the ambiguity surrounding the neutral class, which should yield a significant and direct improvement in neutral-class recall.

Following this, **Phase 3: Structural Optimization with Ordinal Regression** can be explored. If the sentiment classes demonstrably follow a logical order, training a dedicated ordinal model can add another layer of structural constraint that may further improve generalization and performance. This model should be benchmarked directly against the best-performing soft-labeled classifier from Phase 2, using the strict dual-metric evaluation protocol. Finally, **Phase 4: Scalable Enhancement with LLMs** serves as a finishing touch. Here, LLMs are used primarily for data augmentation—generating synthetic positive and negative examples to further balance the dataset and solidify the class boundaries. This is a much safer application than trying to re-label the entire original corpus. Throughout this entire workflow, the evaluation protocol must be strictly adhered to. The primary metrics are Overall Accuracy (>96%) and Neutral-Class Recall. Secondary metrics such as the macro-F1 score, Precision-Recall curves, and the F1-score specifically for the neutral class should be monitored to gain a comprehensive understanding of the trade-offs being made [30] [60]. This systematic, phased approach allows for a thorough investigation of each technique's contribution, ensuring that improvements in neutral-class recall are achieved without compromising the high standard of overall accuracy.

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